

Karthik Chandran Lakshmanasamy

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EXPERIENCE

Principal Software Engineer in Test

MicroStrategy, Tysons, VA

May 2023 – Present

- Developed **end-to-end automation frameworks** using **WebdriverIO** and **Python** for Strategy Mosaic, improving **regression coverage**, test efficiency, and release confidence..
- Built a framework to validate Mosaic's **Universal Semantic Layer architecture**, including semantic models, metrics, relationships, **query generation**, and **SQL accuracy** across enterprise data sources and **BI platforms**.
- Automated **AI-based data modeling** workflows, including linked models, base and derived metrics, automatic metric generation, and **AI-powered analytics**.
- Developed automated testing for Mosaic's **Model Context Protocol** integration with Claude, validating connector setup, tool discovery, **semantic-model access**, **query execution**, **response accuracy**, **permissions**, and **production database or in-memory execution**.
- Validated **LLM-generated** responses against Mosaic's **governed semantic layer** to ensure deterministic, traceable, and accurate business results.
- Implemented **predictive testing** using **TensorFlow** to validate **Retrieval-Augmented Generation** features and improve conversational **AI accuracy for forecast-related queries**.
- Built a Python automation framework from scratch to test customer-managed cloud deployments across **AWS, Azure, GCP, OpenShift**, and **StackIT**.
- Automated **cloud provisioning**, microservice health, configuration, upgrades, autoscaling, workload distribution, concurrency, recovery, and system stability under varying traffic volumes.
- Automated pre and post-deployment microservice checks using **AWS CodeBuild**, validating AI services, **SQL Engine** functionality, and distributed-system components.
- Designed **Jenkins CI/CD** workflows for automated functional, integration, deployment, and regression testing.
- Built a **Python concurrency-testing framework** for event-driven data models to identify throughput, latency, and scalability bottlenecks.
- **Mentored junior engineers** on automation architecture, cloud testing, and tooling selection, helping achieve 90% test coverage across critical components.

Senior Software Engineer in Test

MicroStrategy, Poland

Jul 2019 – May 2023

- Designed an automated framework using **Protractor**, **JavaScript**, and **Cucumber**, cutting manual testing effort for the SAP S/4 Connector, Cloud based application and services.
- Enhanced **Kafka-Flink data stream** performance by 38% by optimizing batch size and processing intervals.
- Played a key role in the certification and release process for **ADLS, S3, and GCS Cloud connectors** in MicroStrategy 2021.
- Led **FedRAMP certification** for the Cloudera **Kubernetes platform**, ensuring compliance with **federal security standards**.
- Built a **Data pipeline** to analyze server logs, providing critical insights into performance metrics for system optimization of backend services.

Software Engineer in Test

MicroStrategy, Tysons, VA

Mar 2018 – Jul 2019

- Designed a framework using **Pytest** and **Selenium** to detect Google BigQuery **data import issues**, enhancing data connectivity.
- Streamlined **CI/CD workflows** by collaborating with **DevOps** to integrate automated tests into pipelines, ensuring seamless deployments and improved release reliability.
- Managed the planning, integration and testing of the **Google BigQuery connector** in MicroStrategy through continuous testing.
- Led the migration from **Teradata** to Google BigQuery, collaborating with cross-functional teams to optimize migration and data accessibility.

Associate Software Engineer in Test

MicroStrategy, Tysons, VA

Mar 2017 – Mar 2018

- Designed a performance testing framework to measure performance of all the big data connectors in MicroStrategy.
- Improved Performance by utilizing **Google storage API** for large scale data import and decreased response time by 44%.
- Lead **Continuous improvement** initiatives to enhance test efficiency, reduce flakiness, and improve defect detection.
- Collaborated with cross-functional teams to define, review, and prioritize requirements and features.
- Implemented and validated **Kerberos authentication** for **Cloudera Data Platform** using Active Directory, enhancing security.

EDUCATION

Master of Science in Computer Science | Texas State University
Bachelor of Engineering in Computer Science | Anna University

Aug 2013 – May 2016
Sep 2009 – May 2013

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Typescript, Java, Bash, Groovy, C++.
- **Databases/Cloud:** BigQuery, Databricks, Snowflake, Oracle, MySQL, MongoDB, PostgreSQL, Redshift, Openshift, StackIT, GCP, AWS, Azure, Redis.
- **Frameworks:** WebdriverIO, Selenium, Playwright, Cypress, Kafka, Node.js, Cucumber, Pytest, JUnit, TestNG, JMeter, Gatling.
- **Tools:** Kubernetes, AWS CodeBuild, Docker, Ansible, Git, Jenkins, Postman, Swagger, Terraform.
- **Machine Learning:** TensorFlow, Pandas, NumPy, Linear Regression models.
- **AI/LLM:** Generative AI, Azure Embedding models, Lang Chain, OpenAI, AWS Bedrock, Gemini, Claude, GPT-OSS, MCP.
- **BI/CRM:** MicroStrategy, Tableau, Qlik, Salesforce, ServiceNow, Power BI, ThoughtSpot, Amazon Quicksight.